

GOURAV KUMAR OJHA

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SUMMARY

AI/ML Researcher with 2 papers under review at Elsevier journals, spanning planetary science (ISRO collaboration) and clinical AI. Experienced in deep learning, NLP, computer vision, multi-modal systems, and large-scale scientific data pipelines. 3x Hackathon Winner and co-founder of a defence-tech AI startup. Passionate about advancing the state of the art through rigorous, reproducible research at the intersection of ML and real-world systems.

PUBLICATIONS (UNDER REVIEW)

- “Interannual Evolution of Martian Dust Activity at Gale Crater: A Multi-Instrument Analysis Using Curiosity Rover and MCS Observations (MY32–MY36).” *Planetary and Space Science, Elsevier* (Under Review, 2026) | In collaboration with Indian Institute of Remote Sensing (ISRO)
- “SETU-DRISHTI: An Explainable Patient-Centric Clinical Decision Support Framework for Early Sepsis Risk Stratification Using Temporal Physiological Features.” *Elsevier* (Under Review, 2026)

EDUCATION

B.Tech in Computer Science (Artificial Intelligence)

Rungta College of Engineering and Technology, Bhilai

Aug 2023 – Present

GPA: 7.8/10

WORK EXPERIENCE

Research Collaborator – Planetary Science & Multi-Instrument Atmospheric Analysis

Aug 2025 – Present

Indian Institute of Remote Sensing (ISRO), Dehradun | Remote

- Designed and implemented a large-scale scientific data pipeline in Python correlating water vapor and dust opacity across multi-instrument datasets (MRO/MCS, Curiosity REMS/SAM, Hubble), processing 10TB+ of planetary observation data; research contributed to a paper submitted to Planetary and Space Science (Elsevier, Under Review)
- Applied ML and statistical modeling techniques for interannual atmospheric analysis and predictive modeling of Martian dust storm propagation across MY32–MY36; conducted reproducible scientific experiments with rigorous validation and quantitative evaluation

Co-Founder & AI Engineer – AstraTech | Defence-Tech Startup | Team of 3

Jan 2026 – Present

Rungta Business Incubation RIU, Bhilai | Remote

- Building an autonomous border intrusion detection system fusing thermal (LWIR), mmWave radar, and seismic sensors with Edge AI for human vs. animal classification; human-in-the-loop authorisation via encrypted LoRaWAN.
- Design patent filed (Patent Pending) for hardware enclosure; currently in active product development phase

RESEARCH PROJECTS

SetuDrishiti – AI-Powered Clinical Decision Support System [[Live](#)]

- Conducted original research on explainable AI for clinical decision support; designed and evaluated a multi-modal deep learning framework for early sepsis risk stratification using temporal physiological features and RAG-based EHR retrieval; research submitted as paper to Elsevier (Under Review)
- Implemented and evaluated Explainable AI methods (SHAP, LIME) for clinician-interpretable model rationale; conducted ablation studies and comparative benchmarking of model variants — Stack: PyTorch, HuggingFace, LangChain, FastAPI, PostgreSQL, Docker

LUMEN AI – Enterprise Financial Assistant [[Github](#)]

- Multi-agentic RAG platform on GCP (Vertex AI, BigQuery) transforming unstructured financial documents into actionable insights with automated risk detection
- Improved financial insight turnaround by 60%; Stack: Python, LangChain, FastAPI, Vector DBs, Docker, CI/CD

Mine-Sigma – Satellite Mining Intelligence System [[Github](#)]

- Developed and evaluated a CNN + YOLO deep learning system for large-scale illegal mining detection from multi-spectral satellite imagery; applied NDVI analysis and geospatial feature engineering to achieve 98% classification accuracy with automated compliance reporting
- Stack: PyTorch, TensorFlow, OpenCV, GeoPandas, Plotly

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL, R

ML / AI: TensorFlow, PyTorch, Scikit-learn, Hugging Face, LangChain, NLP, Natural Language Understanding, Computer Vision, Deep Learning, Multi-Modal Learning, Explainable AI, Statistical Modeling, Scientific Computing (NumPy, SciPy, Pandas), OpenCV, YOLO, SHAP, LIME

Cloud & DevOps: GCP (Vertex AI, BigQuery, Cloud Functions), Docker, CI/CD, Git, MLOps

Web & DB: FastAPI, Flask, React, Next.js, PostgreSQL, MongoDB, Firebase, Vector DBs

Blockchain: Solidity, Ethereum, Hardhat, Web3.js

CERTIFICATIONS

- Google Cloud Skills Boost – GCP Fundamentals & Applied ML (2025)
- Microsoft Azure Fundamentals (AZ-900) | AICTE AI & ML Fundamentals | Postman API Expert
- Research Intern – Supernova Cosmology Project, India Space Week (Jun–Jul 2025): Estimated Hubble Constant from JWST/HST Type Ia Supernovae data; published findings on Hubble tension
- Open Source Contributor, GSSoC 2024 (Oct–Nov 2024): ML modules across multiple repos; 15–20% performance improvements

ACHIEVEMENTS

- 1st Place – E-Summit Startup Competition | Vyom Cultural Fest, RISU**
 - Won top prize pitching AstraTech in Shark Tank-style competition; awarded by Padma Shri Anand Kumar

March 2026
- 1st Place – Hack-Sprint Hackathon | GDG on Campus RCET & Infinity Coders**
 - Led Team Alien X to build AI-powered Real-time Sepsis & Deterioration Monitoring System in a 12-hour sprint; integrated real-time patient vitals with ML-based alert systems

March 2026
- 3x Regional AI Hackathon Winner**

2024–present
- 1x Project Show Case Winner**

August 2025